
SPATIAL ANALYSIS OF FLOOD POTENTIAL, AFFECTED FACILITIES, AND EVACUATION ROUTING IN THE TALIAOKENG RIVER BASIN

1. INTRODUCTION

In recent years, the increasing frequency and intensity of extreme rainfall events caused by climate change have significantly elevated flood risks in urban areas. The Taliaokeng River Basin, located in Xinzhuang District, New Taipei City, has long been recognized as a flood-prone area due to extensive upstream development, low-lying terrain, and limited drainage capacity. Although a series of flood mitigation projects have been implemented by governmental agencies to reduce flood hazards, certain sections of the basin remain highly vulnerable under extreme rainfall conditions.

The basin contains numerous Vulnerable Facilities, including hospitals, nursing homes, elderly welfare institutions, long-term care centers, kindergartens, and infant care centers. These facilities accommodate populations with limited mobility, reduced self-evacuation capability, or special care needs. Once flooding occurs, delayed evacuation may result in severe threats to human safety and emergency response operations.

Therefore, this study applies Geographic Information System (GIS) techniques to evaluate flood risks affecting Vulnerable Facilities within the Taliaokeng River Basin, assess the accessibility of Emergency Shelters, and develop safe evacuation routes under different flood scenarios. The results are expected to support disaster prevention planning, emergency response, and resource allocation for local authorities.

2. STUDY AREA

The study area is located within the Taliaokeng River Basin in New Taipei City, Taiwan. To comprehensively evaluate flood impacts and evacuation accessibility, a 300-meter buffer zone was established along both sides of the main channel of the Taliaokeng River. This buffer area was defined as the spatial boundary of the study area, as illustrated in **Figure 1**.

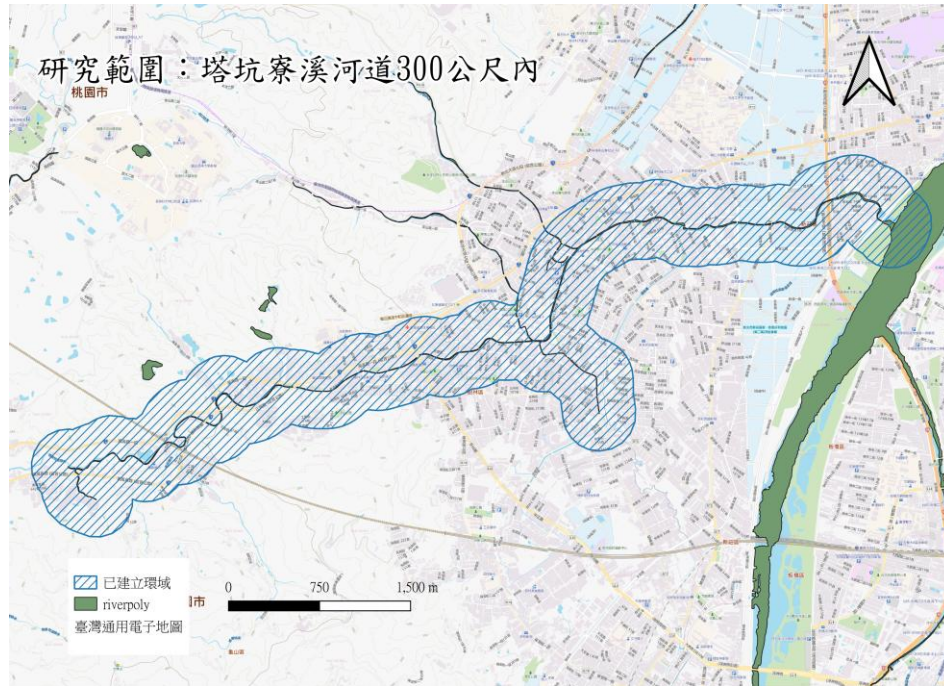


Figure 1. Study Area of the Taliaokeng River Basin

Facilities within the study area were classified into two categories: Emergency Shelters and Vulnerable Facilities.

Emergency Shelters consist of disaster prevention parks, school evacuation sites, community activity centers, and officially designated emergency evacuation shelters. These facilities provide temporary accommodation and emergency support during flood events. The spatial distribution of Emergency Shelters is presented in Figure 2.

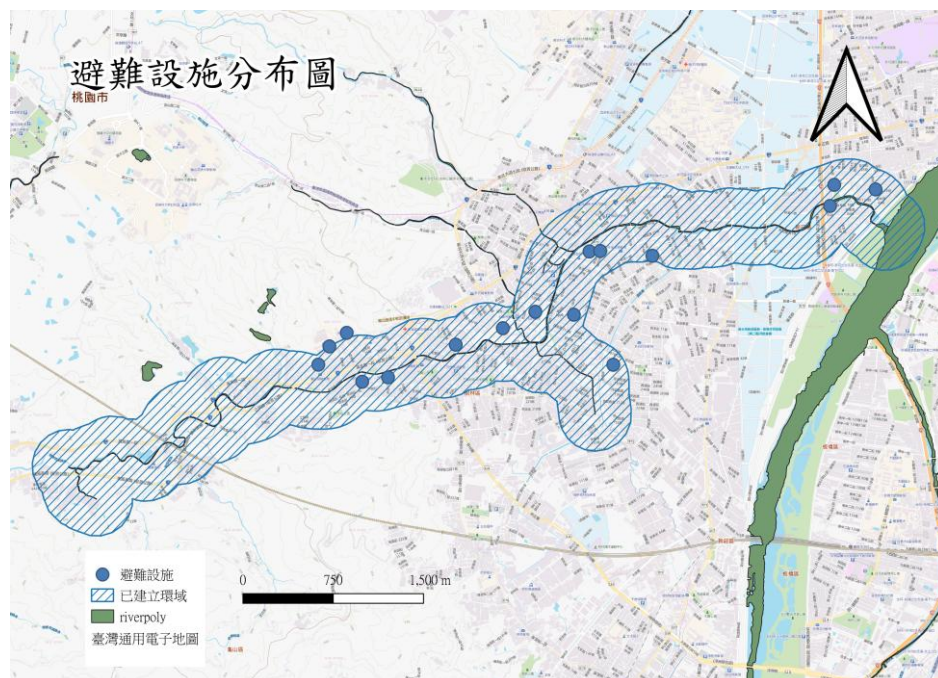


Figure 2. Spatial Distribution of Emergency Shelters

Vulnerable Facilities include hospitals, nursing homes, elderly welfare institutions, long-term care centers, kindergartens, and infant care centers. These facilities accommodate populations with limited mobility or reduced self-evacuation capability, making them particularly susceptible to flood disasters. Therefore, ensuring their accessibility to emergency shelters is a critical component of disaster management planning. The spatial distribution of Vulnerable Facilities is shown in **Figure 3**.

A total of 36 Vulnerable Facilities were identified within the study area and selected as the primary targets for subsequent flood risk assessment and evacuation route analysis.

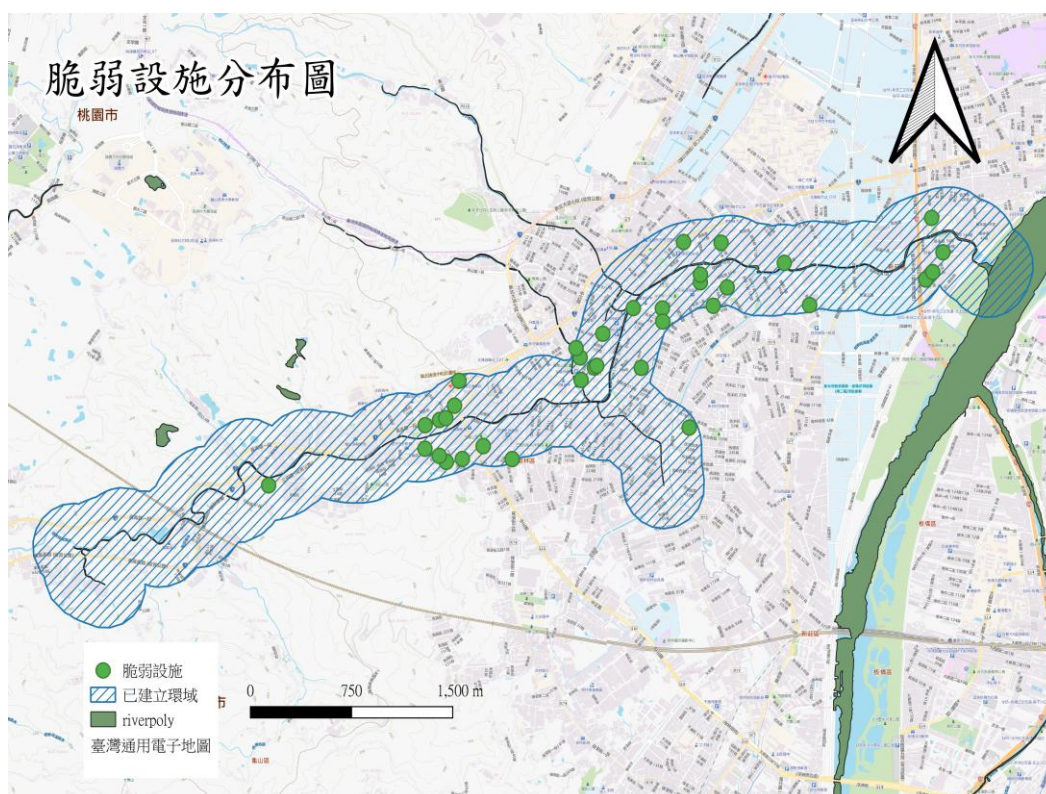


Figure 3. Spatial Distribution of Vulnerable Facilities

3. DATA COLLECTION

3.1 Flood Potential Data

Flood potential maps were obtained from the Flood Potential Map Database provided by the Water Resources Agency (WRA), Ministry of Economic Affairs, Taiwan. To represent flood conditions under different rainfall durations and precipitation intensities, ten flood scenarios were collected, including 6-hour, 12-hour, and 24-hour rainfall events with accumulated rainfall ranging from 150 mm to 650 mm. Each dataset contains estimated flood depths under a specific rainfall scenario and was used to evaluate the impacts of flooding on facilities and transportation accessibility within the study area.

3.2 Facility Data

Facility data were collected from multiple open datasets provided by the Taiwan Government Open Data Platform and local government agencies. Additional facility information not available in official datasets was supplemented through Google Maps verification and manual geocoding to improve data completeness and positional accuracy.

The collected facilities were categorized into two groups: Emergency Shelters and Vulnerable Facilities. Emergency Shelters include schools, community activity centers, and officially designated emergency evacuation shelters. Vulnerable Facilities include hospitals, nursing homes, elderly welfare institutions, long-term care centers, kindergartens, and infant care centers.

After data integration and screening, a total of 36 Vulnerable Facilities were identified within the study area. All facility locations were converted into GIS point layers and projected to the TWD97 / TM2 coordinate system (EPSG:3826) for subsequent spatial and network analyses.

4. METHODS

An overview of the research framework and data processing procedures used in this study is presented in **Figure 4**. The methodology consists of five major stages: (1) vulnerable facility risk quantification, (2) shelter availability screening, (3) isolated vulnerable facilities identification, (4) flood-affected road network construction and dynamic flood-island identification, and (5) evacuation route planning.

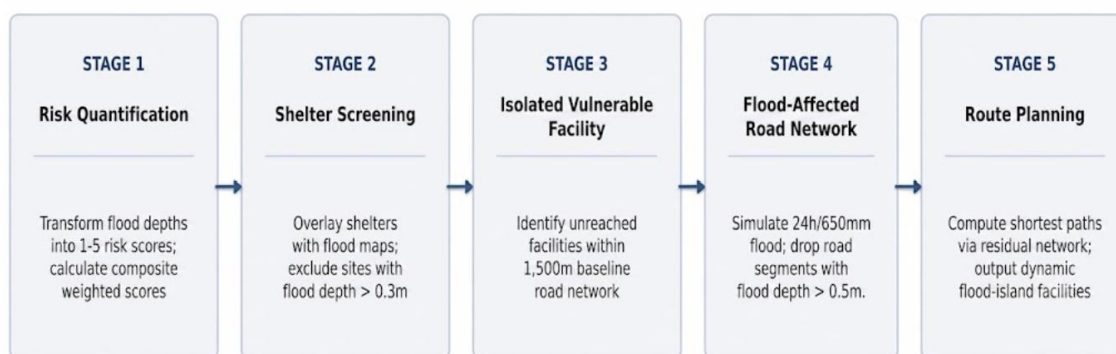


Figure 4. Research Framework and Methodology Flowchart

4.1 Vulnerable Facility Data Processing and Risk Quantification

To systematically evaluate the flood risks across the 36 identified vulnerable facilities under various precipitation intensities, a spatial data processing and risk-scoring framework was established. **This framework references the disaster risk assessment methodology utilized by the National Disasters Prevention and Protection Center (NCDR)**, which aligns with the IPCC SREX report defining Risk as the intersection of Hazard, Vulnerability, and Exposure (as illustrated in **Figure 5**). Specifically, **this study evaluates the intersection of two localized core indicators: "Hazard-Probability" and "Vulnerability-Inundation Depth"**. Hazard represents the climatological probability of historical and future extreme rainfall scenarios, whereas Vulnerability reflects the physical disruption and operational impacts imposed on

facilities under specific flood depths. This framework standardizes qualitative spatial inundation intervals into a single mathematical proxy to prioritize facility hazard levels through three sequential steps:

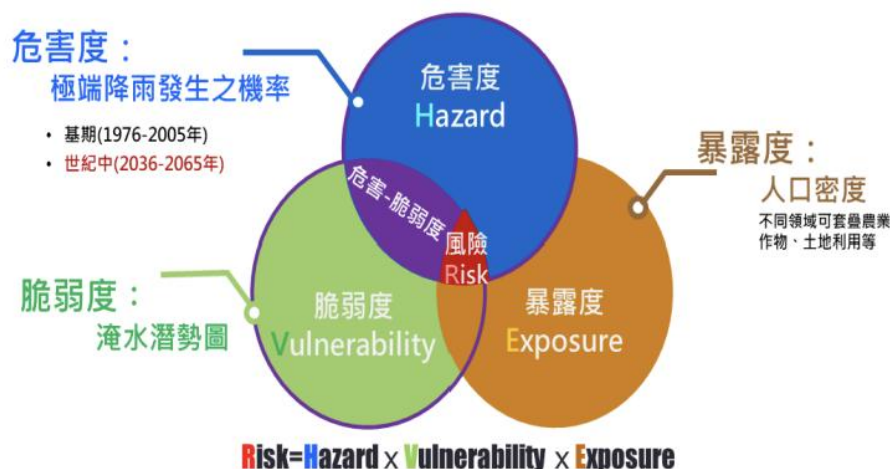


Figure 5. Indicators for Climate Change-Induced Flood Risk Maps

Step 1: Flood Depth Transformation

Through spatial overlay analysis between the facility point layers and the ten flood potential maps, the qualitative or raw inundation depth intervals at each facility location were extracted. These depths were converted into numeric Risk Scores ranging from 1 to 5 to represent the physical severity of transport and evacuation disruptions, as defined in **Table 1**:

The five-tier inundation depth classification criteria deployed in Table 1 reference the quantitative evaluation logic for "flood depth vulnerability indicators" published by the National Disasters Prevention and Protection Center (NCDP). **Within NCDP's national-level disaster risk governance framework, inundation depths are systematically divided into five distinct intervals**; we have subsequently converted these intervals into corresponding weights based on their operational impact severity. By adopting this national-level indicator configuration, this study ensures empirical rigor and localized policy alignment within the context of Taiwan's climate change adaptation planning.

Table 1. Flood Depth Risk Scoring and Impact Matrix

Inundation Depth (m)	Risk Score (S)	Operational Impact & Description
0.0 ~ 0.3	1	Minor flooding: Minimal impact on normal operations and pedestrian movement.
0.3 ~ 0.5	2	Water accumulation: May affect the passage of low-clearance vehicles; pedestrian speed reduces.
0.5 ~ 1.0	3	Knee-level flooding: Passage for individuals with limited mobility is completely obstructed.

Inundation Depth (m)	Risk Score (S)	Operational Impact & Description
1.0 ~ 2.0	4	Thigh-to-waist flooding: Substantial hazard; vehicle movement is impossible; high danger level.
2.0 ~ 3.0	5	Extreme flooding: Water depth exceeds human height; critical danger; immediate vertical rescue required.

Step 2: Scenario Weighting

The ten multi-scenario rainfall datasets provided by the Water Resources Agency (WRA) represent varying return periods and durations (6-hour, 12-hour, and 24-hour events). To reflect their long-term climatological probabilities, these scenarios were categorized into four groups and assigned statistical weights based on their historical occurrence frequencies, as outlined in **Table 2**:

Based on the data from the Xinzhuang Rainfall Warning Station provided by the Water Resources Agency (WRA), **the 6h_150mm scenario has officially reached the station's Tier 1 warning threshold**. This indicates that the scenario carries a high probability of actual occurrence and substantial hazard risks in the real-world environment (as illustrated in **Figure 6**). Belonging to a high-frequency disaster context that demands critical attention, it is classified under the primary assessment weight group (S_{high}) with the highest assigned weight of 0.4.

Conversely, while other higher-intensity cumulative rainfall scenarios (such as 24h_650mm) can more comprehensively reflect the failure states and ultimate vulnerability of the urban drainage system and vulnerable facilities under extreme climate conditions, their statistical return periods become significantly longer as the precipitation values surpass the baseline thresholds, resulting in a lower frequency of occurrence. Therefore, this study sequentially decreases the weights for these extreme scenarios.

Table 2. Rainfall Scenario Weights and Probability Matrix

Rainfall Scenario	Included Rainfall Scenarios	Weight	Occurrence Frequency Description
S_{high}	6h_150mm, 12h_200mm, 24h_200mm	0.4	High-frequency event; highly likely to occur.
$S_{\text{med-high}}$	6h_250mm, 12h_300mm, 24h_350mm	0.3	Medium-high frequency event.
S_{low}	6h_350mm, 12h_400mm, 24h_500mm	0.2	Low-frequency event.

Rainfall Scenario	Included Rainfall Scenarios	Weight	Occurrence Frequency Description
S_{extreme}	24h_650mm	0.1	Extreme climate event; lowest probability.

警戒雨量站			
● 新莊			
未達警戒			
影響範圍:新莊區-思源里,化成里,營盤里,福基里,頭前里,中原里,立功里,文聖里,文明里,仁愛里,立基里,榮和里,國泰里,海山里,全安里,文衡里,文德里,忠孝里,中港路,新北大道,中原路,幸福路,復興路,化成路,思源路,中華路			
時間	雨量	警戒	
		二級	一級
1H	0	40	50
3H	0	90	100
6H	0	130	150

Figure 6. WRA Rainfall Warning Classifications and Xinzhuang Station Alert Thresholds

Step 3: Composite Risk Calculation

To derive a unified metric for ranking the comprehensive vulnerability of each facility, a composite **Weighted Flood Risk Score** was computed. The score aggregates the transformed risk values across all 10 scenarios multiplied by their respective group probabilities, formulated as follows:

$$R_{\text{weighted}} = 0.4 \times \sum S_{\text{high}} + 0.3 \times \sum S_{\text{med-high}} + 0.2 \times \sum S_{\text{low}} + 0.1 \times S_{\text{extreme}}$$

Where sum denotes the sum of the transformed risk scores within that specific frequency group, facilities were subsequently ranked based on their Weighted Flood Risk Score values to assist emergency planners in identifying top-priority mitigation sites.

4.2 Shelter Availability Screening

Prior to conducting network analysis, the availability of Emergency Shelters under flood conditions was evaluated. Flood depth information from each flood scenario was overlaid with the shelter locations using spatial overlay analysis.

Considering that flood depths exceeding 0.3 m may significantly hinder pedestrian evacuation and emergency vehicle access, shelters located within areas with flood depths greater than 0.3 m were excluded

from further analysis. After screening, seven Emergency Shelters remained available and were selected as the evacuation destinations for subsequent network analyses.

The spatial distribution of the selected Emergency Shelters is shown in **Figure 7**.

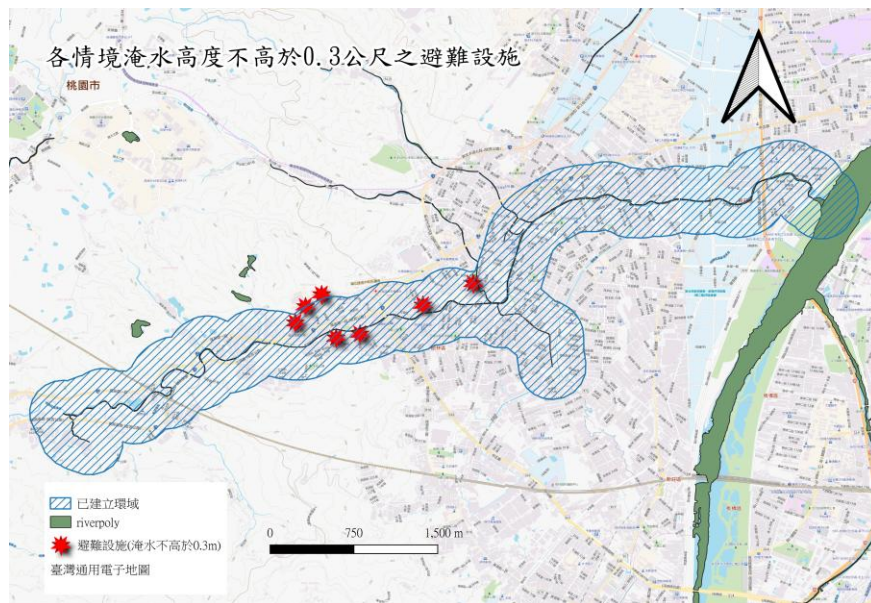


Figure 7. Available Emergency Shelters under Flood Scenarios (Flood Depth ≤ 0.3 m)

4.3 Isolated Vulnerable Facilities Identification

The first stage of network analysis established a baseline scenario to evaluate the geographical equity of resource distribution under normal, unflooded conditions. This step aimed to uncover pre-existing spatial gaps between Vulnerable Facilities and active Emergency Shelters without accounting for climate-induced road disruptions.

Network modeling was executed via the Service Area (from Layer) algorithm embedded in the QGIS Network Analysis toolbox. All geospatial vector data—including the road network and facility layers—were projected to the TWD97 / TM2 coordinate system (EPSG:3826) to ensure precise metric distance metrics along the network arcs.

Using the 7 screened safe shelters as network origins, a maximum service distance threshold of 1,500 meters was mapped outward along the continuous road network. This distance threshold was adopted based on the urban disaster prevention planning guidelines published by the Architecture and Building Research Institute (ABRI), Ministry of the Interior, and was used here as an analytical parameter rather than an official classification of accessibility failure. Any Vulnerable Facility whose spatial coordinates failed to intersect with the generated 1,500-meter service area polygons was mathematically classified as an Isolated Vulnerable Facility. As illustrated in **Figure 8**, a total of eight facilities were identified within this category. This identification indicates areas characterized by inherent institutional supply deficits that require long-term infrastructure expansion or localized shelter deployment.

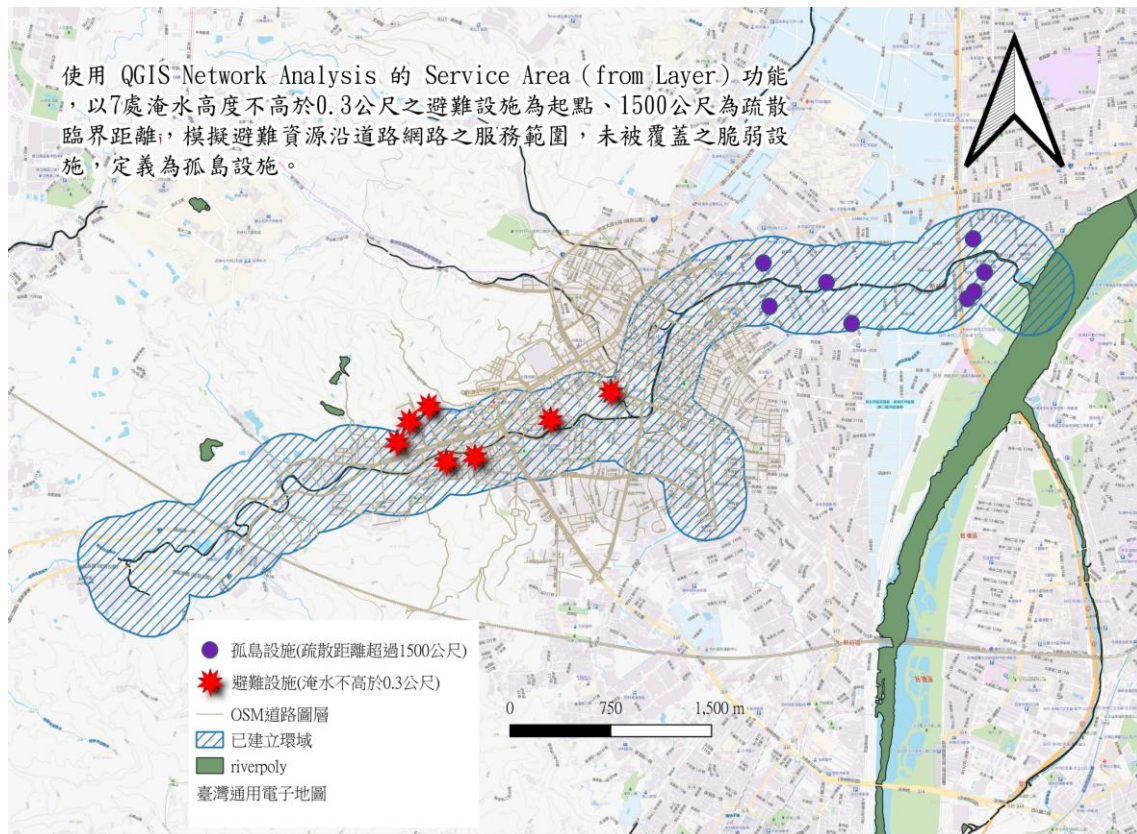


Figure 8. Isolated Vulnerable Facilities Identified by Service Area Analysis

4.4 Flood-Affected Road Network Construction

The second stage evaluated evacuation accessibility under an extreme flood scenario: the **24-hour accumulated 650-mm rainfall event**. This scenario was selected to represent severe flooding conditions. To construct the flood-affected road network, the road network data was intersected with the 24h-650mm flood potential map to assign a flood-depth attribute to each road segment. In terms of transportation safety, a flood depth greater than **0.5 meters** is the absolute physical limit for standard vehicles and emergency trucks, as this depth typically exceeds half the height of a standard vehicle tire. Therefore, road segments with flood depths exceeding 0.5 meters were considered impassable and removed from the analysis. As shown in **Figure 9**, the remaining usable roads formed the **Residual Road Network**, representing the available routes for general traffic and emergency rescue operations during the flood event.

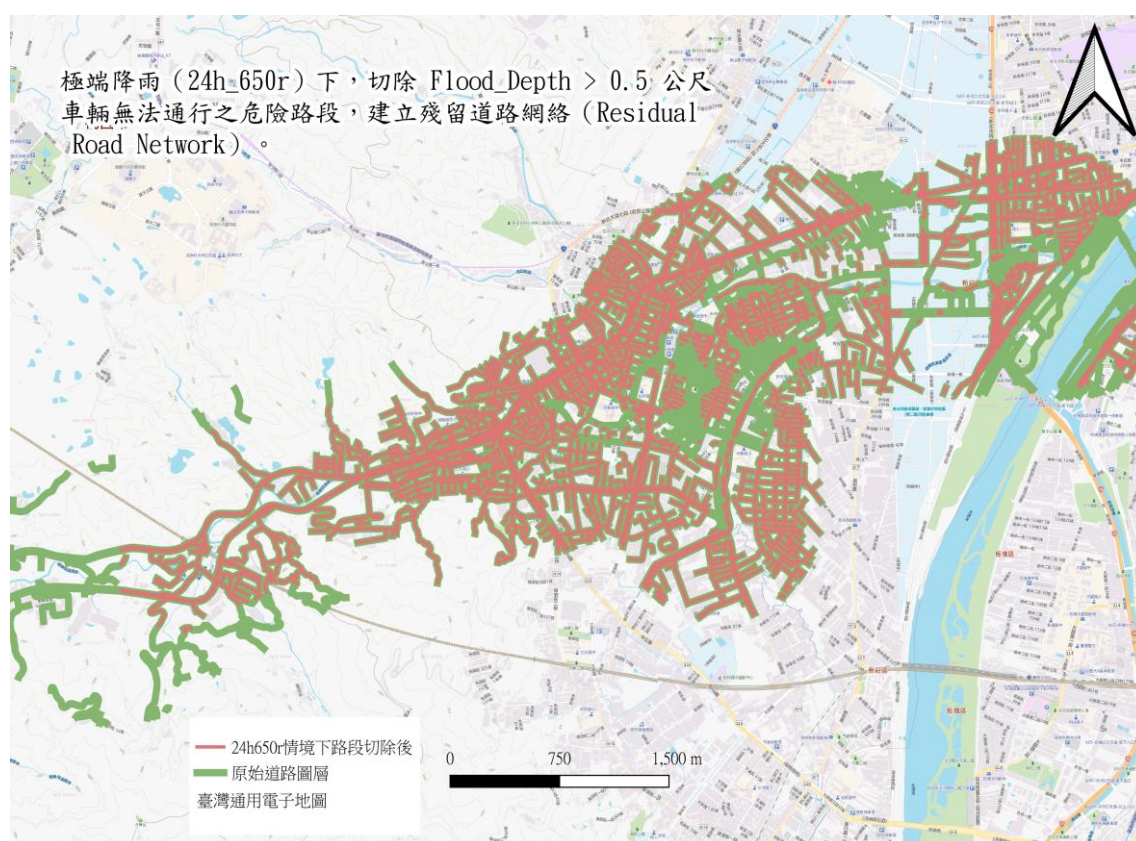


Figure 9. Full and Residual Road Network under Flood Scenario

4.5 Evacuation Route Planning

Utilizing the compiled Residual Road Network, this stage deployed the most stable **Network Analysis** toolbox built into QGIS to calculate all accessible routes and their corresponding metric distances from each Vulnerable Facility to the remaining safe shelters. Specifically, the **Shortest Path (Point to Layer)** algorithm was executed to model the spatial connectivity under the designated rainfall scenario. Rather than simply finding a single destination, the algorithm mapped all mathematically possible non-flooded (< 0.5 m) corridors for each facility. By exporting these routing results into a distance matrix spreadsheet, a structured multi-destination proximity evaluation was conducted:

1. **Optimal Destination Selection:** For each connected facility, the system extracted and compared the cumulative real-world travel distances across all reachable safe shelters. The route registering the absolute shortest travel distance was selected as the **Optimal Evacuation Route**, successfully minimizing the population's exposure time to risks during transit. As demonstrated in the case study of 艾倫戴爾幼兒園 (Figure 10), the system computed all viable routes to the surrounding shelters. The quantitative comparison of these distances is summarized in Table 3, where 迴龍市民活動中心 was selected as the optimal destination due to its minimum accessible distance of 2,387.15 meters.
2. **Inaccessible Facility Filtration:** If the routing engine encountered network disconnections caused by surrounding road segments exceeding the 0.5-meter flood depth threshold, the connection to that specific shelter was automatically voided. For example, 丹鳳高級中學 was filtered out as an inaccessible shelter for the case facility due to complete road network isolation.

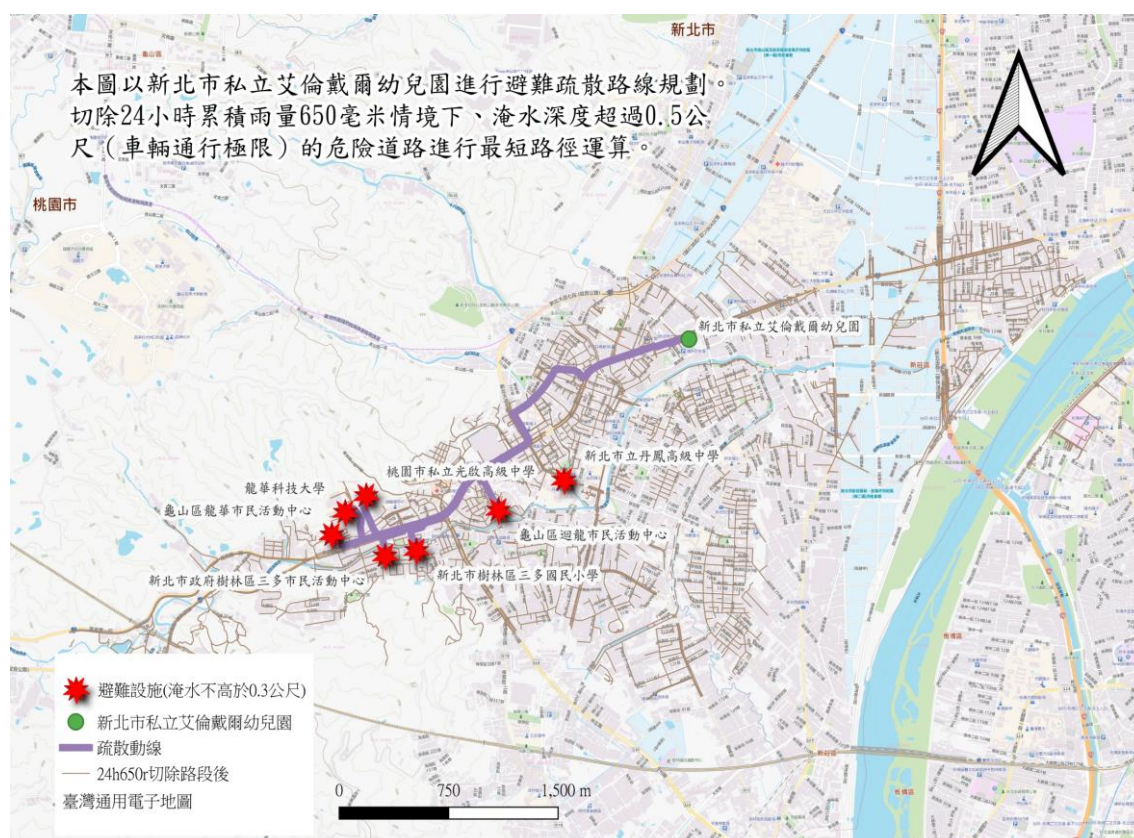


Figure 10. Simulation of Accessible Shelters and Evacuation Routes

Table 3. Comparison of Evacuation Distances and Accessibility for Case Facility

Destination	龍華科技大學	光啟高級中學	三多國民小學	丹鳳高級中學	迴龍市民活動中心	樹林區三多市民活動中心	龜山區龍華市民活動中心
艾倫戴爾幼兒園	3310.09 m	3172.61 m	2913.03 m	—	2387.15 m	3031.39 m	3294.53 m

(Note: The comprehensive evaluation distance matrix covering all 19 vulnerable facilities within the study area is provided in Appendix A.)

(Note: The evacuation route maps and corresponding routing results for all 28 vulnerable facilities are presented in Appendix B.)

5. RESULTS AND DISCUSSIONS

5.1 Flood Risk Assessment of Vulnerable Facilities

The flood risk assessment results reveal considerable differences in hazard vulnerability among the 36 analyzed vulnerable facilities. Based on the Weighted Flood Risk Scores calculated from the ten multi-scenario rainfall simulations, the prioritized risk ranking is visualized in Figure 11.

The spatial overlay and weight computations identified the top three highest-risk facilities within the study area:

1. 市立光華國小附設幼兒園 registered the highest composite score of 9.7, reflecting its catastrophic exposure across almost all extreme precipitation thresholds.
2. 新北市私立艾倫戴爾幼兒園 ranked second with a composite score of 7.1.
3. 新北市新莊裕民公共托育中心 and 新北市立新莊幼兒園中心分班 tied for the third position, both obtaining a score of 5.4.

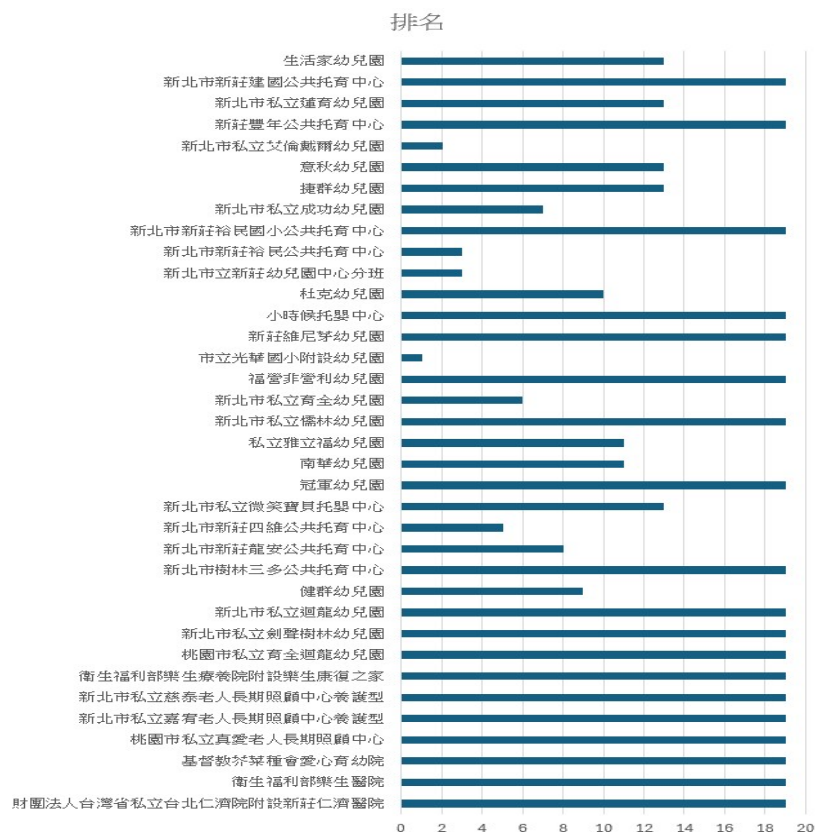


Figure 11. Risk Ranking and Weighted Flood Risk Scores of Vulnerable Facilities

These high-scoring facilities exhibit severe exposure to multiple inundation scenarios and must be prioritized by local authorities as critical focal points for immediate disaster preparedness, mitigation funding, and emergency response planning.

5.2 Analysis of 8 Isolated Vulnerable Facilities

The 7 operationally available Emergency Shelters were mapped to simulate a 1,500-meter service area along the road network. The spatial analysis successfully identified **8 isolated vulnerable facilities** that fall completely outside this standard institutional resource coverage, which include:

- 新莊豐年公共托育中心
- 新北市立新莊幼兒園中心分班
- 意秋幼兒園
- 捷群幼兒園
- 新北市新莊建國公共托育中心
- 生活家幼兒園
- 新北市私立蓮育幼兒園
- 財團法人台灣省私立台北仁濟院附設新莊仁濟醫院

5.3 Evacuation Accessibility Analysis

For the remaining vulnerable facilities located within the service area, the nearest Emergency Shelter and evacuation distance were calculated. The results indicate that evacuation accessibility varies among facilities, reflecting differences in the spatial distribution of shelter resources and road network connectivity within the study area. Under normal conditions, emergency planners must pay attention to these inherent spatial disparities, as institutions with longer baseline distances require more rigorous early-stage logistics and higher transportation readiness even before any heavy rainfall begins.

5.4 Analysis of 19 Facilities with Navigable Routes

After removing road segments with flood depths exceeding 0.5 m, which represents the absolute physical limit for standard vehicles and emergency trucks, the evacuation network under the 24-hour accumulated 650-mm rainfall scenario was evaluated. Through the spreadsheet distance matrix and spatial routing optimization, the network analysis successfully generated alternative evacuation routes for 19 vulnerable facilities that managed to maintain continuous physical connection to safety nodes.

Although their primary shortest paths were disrupted by the flood, the routing engine identified cost-minimizing trajectories through the remaining usable road segments to guide them to safe destinations like 迴龍市民活動中心.

Rerouting these 19 accessible institutions through non-flooded corridors effectively minimizes the population's transit time and hazard exposure during a crisis.

Planning & Cautionary Measures: For these 19 facilities, authorities must advance road clearing priorities and ensure that dynamic digital routing information can be pushed to facility staff in real time. Since these routes rely on alternative local streets, keeping these specific backup corridors free from traffic congestion or secondary hazards (like fallen trees or gridlock) is the most critical operational prerequisite during an emergency.

5.5 Analysis of 9 Facilities Trapped as Flood-Islands

Conversely, severe flood disruptions triggered complete network degradation in specific sectors under the extreme rainfall scenario. The analysis precisely identified **9 vulnerable facilities** that became completely cut off, registering zero valid evacuation routes because their surrounding road infrastructure comprehensively exceeded the 0.5 m inundation threshold.

To ensure complete empirical transparency, the 9 institutional hot spots trapped as flood-islands are comprehensively identified as follows:

- 杜克幼兒園
- 小時候托嬰中心
- 新北市私立微笑寶貝托嬰中心
- 新莊維尼芽幼兒園
- 福營非營利幼兒園
- 新北市立新莊四維公共托育中心
- 新北市私立成功幼兒園
- 市立光華國小附設幼兒園
- 健群幼兒園

Standard horizontal road evacuation and emergency vehicular rescue operations are physically impossible for these 9 trapped locations during peak flooding.

Planning & Cautionary Measures: For these 9 isolated vulnerable facilities, standard horizontal road evacuation is physically impossible once flooding starts. Therefore, authorities and facility administrators must pre-arrange specialized early evacuation protocols (such as executing mandatory closure and institutional clearance hours before peak precipitation occurs) or verify building structural safety for vertical in-place sheltering, ensuring that high-floor spaces are equipped with sufficient emergency power, medical supplies, and rations.

5.5 Discussion

The results demonstrate that integrating flood potential maps with GIS network analysis can effectively identify high-risk vulnerable facilities, isolated vulnerable facilities, and evacuation accessibility issues. The proposed approach provides useful information for emergency planning, shelter allocation, and flood disaster management in the Taliaokeng River Basin.

To further evaluate the practicality of emergency response strategies, future analysis may incorporate 3D building models provided by the National Land Surveying and Mapping Center (NLSC). For facilities identified as flood-islands, building height and floor-level information can be used to assess the feasibility of vertical evacuation. Facilities with sufficient upper-floor space may adopt in-place vertical sheltering during extreme flood events, reducing the need for immediate horizontal evacuation and improving emergency management flexibility.

6. CONCLUSION

This study integrated flood potential maps, GIS spatial analysis, and network routing techniques to evaluate the multi-scenario flood risks and evacuation accessibility of 36 vulnerable facilities within the Taliakeng River Basin. The experimental analysis successfully identified seven operationally available Emergency Shelters, unmasked 8 isolated vulnerable facilities under normal conditions, and revealed critical differences in flood vulnerability across the institutions. Childcare-related facilities generally exhibited significantly higher flood risk scores than hospitals and elderly care institutions. Furthermore, routing analysis under the 24-hour 650-mm rainfall scenario demonstrated that flood-induced road disruptions drastically degrade evacuation accessibility, trapping 9 facilities as flood-islands.

6.1 Alignment with Sustainable Development Goals (SDGs)

By transforming static Points of Interest (POIs) and qualitative flood maps into dynamic evacuation strategies, this project bridges the gap between spatial data and local climate adaptation, directly contributing to four United Nations SDGs:

- **SDG 3: Good Health and Well-Being:** The framework helps local emergency operations save more lives during extreme weather events and ensures healthcare continuity for highly vulnerable, mobility-restricted populations.
- **SDG 10: Reduced Inequalities:** By prioritizing pre-flood resources for the highest-risk facilities and bridging the geographical information gap, the system prevents spatial inequity in municipal disaster resource allocation.
- **SDG 11: Sustainable Cities and Communities:** The dynamic network analysis provides empirical tools for optimizing public safety infrastructure and enhancing overall urban resilience against hydro-meteorological hazards.

6.2 Future Outlook

While the proposed framework provides actionable, empirical data for municipal governance, certain directions remain open for future enrichment:

1. **Incorporation of Shelter Capacity Constraints:** Future studies can integrate the maximum accommodation capacities and medical supply thresholds of each designated shelter into the routing engine to avoid overwhelming a single destination.
2. **Dynamic Traffic and Obstruction Modeling:** Incorporating real-time vehicular traffic flows, pedestrian evacuation speeds, and micro-scale topographic barriers would enhance the behavioral accuracy of the routing models.
3. **Integration of Building Floor Levels for Vertical Evacuation:** Future research can incorporate building structural data—specifically the number of floor levels and vertical space availability of each vulnerable facility. This enables emergency planners to evaluate whether a facility trapped as a flood-island can execute safe "vertical in-place sheltering" to upper floors when road-based horizontal evacuation becomes physically impossible, thereby optimizing customized emergency protocols.

7. Appendix

1. Data collection source

- 經濟部水利署臺灣水災潛勢風險圖資應用服務平台（2026）：
<https://fhy.wra.gov.tw/fhyv2/disaster/downloads>
- 國民小學名錄（2014）：<https://data.gov.tw/dataset/6087>
- 國民中學名錄（2014）：<https://data.gov.tw/dataset/6088>
- 幼兒園名錄（2014）：<https://data.gov.tw/dataset/6086>
- 社家署全球資訊網 全國機構名冊（2026）：<https://www.sfaa.gov.tw/sfaa/list/detail/5fW/5pe>
- 新北市私立托嬰機構名冊（2022）：<https://data.ntpc.gov.tw/datasets/69cecdb0-7796-48df-84e5-99e4f1274245>
- 衛生福利部食品藥物管理署 全台醫院分級名單（2014）：
<https://www.fda.gov.tw/tc/includes/GetFile.ashx?id=f636694268478889043>

2. References

- 新北市新莊區警戒雨量資訊（2026）：<https://fhy.wra.gov.tw/fhyv2/service/alertQuery>
- 國家災害防救科技中心氣候變遷組（2020）。《考量多模式以及不同空間尺度應用之氣候變遷淹水風險圖》：<https://www.ncdr.nat.gov.tw/UploadFile/Newsletter/20201222161942.pdf>
- 內政部建築研究所（2015）。《都市防災規劃調適與變遷因應指南》：
https://www.abri.gov.tw/News_Content_Table.aspx?n=807&s=38418

3. Appendix

- Appendix A. Comprehensive Evacuation Distance Matrix for All Vulnerable Facilities

Optimal Destination	龍華科技大學 (m)	光啟高級中學 (m)	三多國民小學 (m)	丹鳳高級中學 (m)	迴龍市民活動中心 (m)	三多市民活動中心 (m)	龍華市民活動中心 (m)	Nearest Vulnerable Facility
新北市私立艾倫戴爾幼兒園	3310.10	3172.61	2913.03	—	2387.15	3031.39	3294.54	龜山區迴龍市民活動中心
新北市私立育全幼兒園	2649.62	2512.13	2250.08	—	1723.12	2368.44	2634.06	龜山區迴龍市民活動中心
新北市新莊裕民國小公共托育中心	2723.33	2585.84	2446.92	—	2183.35	2204.22	2707.77	龜山區迴龍市民活動中心

新北市新莊龍安公共托育中心	2723.33	2585.84	2446.92	—	2183.35	2204.22	2707.77	龜山區迴龍市民活動中心
新北市私立儒林幼兒園	2767.98	2630.49	2368.44	—	1841.48	2486.80	2752.42	龜山區迴龍市民活動中心
新北市新莊裕民公共托育中心	2713.01	2575.52	2313.47	—	1786.51	2431.83	2697.45	龜山區迴龍市民活動中心
衛生福利部樂生醫院	1139.84	951.34	743.47	—	620.81	861.69	1124.28	龜山區迴龍市民活動中心
衛生福利部樂生療養院附設樂生康復之家	884.18	746.70	487.81	—	505.35	606.03	868.63	三多國民小學
南華幼兒園	794.28	656.79	347.20	—	555.85	467.55	778.72	三多國民小學
桃園市私立真愛老人長期照顧中心	746.73	609.25	299.65	—	603.40	420.00	731.18	三多國民小學
桃園市私立育全迴龍幼兒園	657.68	520.20	178.37	—	724.68	330.95	642.12	三多國民小學
基督教芥菜種會愛心育幼院	1222.60	1309.51	1318.16	—	1953.52	1140.06	801.51	龍華市民活動中心
新北市樹林三多公共托育中心	744.18	606.69	467.77	—	1210.72	225.08	728.62	三多市民活動中心
私立雅立福幼兒園	1003.53	866.04	727.12	—	1130.21	484.42	987.97	三多市民活動中心
新北市私立劍聲樹林幼兒園	982.82	845.33	706.41	—	1109.50	463.72	967.26	三多市民活動中心
冠軍幼兒園	1271.36	1133.87	994.95	—	815.54	752.25	1255.80	三多市民活動中心
新北市私立嘉宥老人長期照顧中心（養護型）	1428.78	1291.29	1071.93	—	717.71	909.67	1413.22	龜山區迴龍市民活動中心

新北市私立慈泰老人長期照顧中心（養護型）	1428.78	1291.29	1071.93	－	717.71	909.67	1413.22	龜山區迴龍市民活動中心
新北市私立迴龍幼兒園	1548.11	1410.62	1271.70	－	1006.50	1029.00	1532.55	龜山區迴龍市民活動中心

● Appendix B. Evacuation Route Maps for All Vulnerable Facilities





